

Intermittent Control Strategy for Synchronization Analysis of Time-Varying Complex Dynamical Networks

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Abstract—The p th moment exponential synchronization (PMES) problem for time-varying complex dynamical networks (CDNs) with multiple time-varying delays (TCDNMTDs) by aperiodically intermittent control is studied in this paper. Compared with related work, the coupling structure of CDNs is time varying. Also, internal time-varying delays and coupling time-varying delays are taken into consideration and they are different from each other. It is worth emphasizing that the intermittent control scheme proposed here is nonperiodical. Some criteria are derived to make TCDNMTDs achieve PMES by using the Lyapunov method and graph theory. Additionally, we also consider the synchronization of second-order Kuramoto oscillators and coupled Chua's circuits and some numerical simulations are presented to verify the validity of our results.

Index Terms—Aperiodically intermittent control (AIC), multiple time-varying delays, p th moment exponential synchronization (PMES), time-varying complex dynamical networks (CDNs).

I. INTRODUCTION

THE SYNCHRONIZATION problem of complex dynamical networks (CDNs) has become a hot topic and it has wide applications in communication networks, image processing, and so on [1]–[9]. When studying the synchronization problem of CDNs, time delays play an important role since time delays can cause unexpected synchronization performances and oscillations in CDNs. Generally, time delays can be classified into internal delays and coupling delays. Internal delays usually occur inside the dynamical nodes while coupling delays affect information exchanging process between dynamical nodes. In particular, coupling delays consist of self-delays and neighboring delays. Self-delays and neighboring delays affect node's own state

and neighbor's state, respectively. Recently, internal delays and coupling delays have been extensively studied by many scholars in [8] and [10]–[12]. Guo *et al.* [8] considered the synchronization of CDNs and proposed their coupling term as $\Gamma(x_h(t - \tau) - x_k(t))$, which means their coupling term is linear and only neighboring delays were taken into account. In [10], the linear coupling term was given in the form of $\Gamma(x_h(t - \tau) - x_k(t - \tau))$, where the self-delay and neighboring delay were considered to be the same constants. In [11], self-delay was different from neighboring delay and they gave their linear coupling term as $\Gamma(x_h(t - \alpha) - x_k(t - \sigma))$, in which the time delays are constant. In real world, however, the switching speeds of amplifiers are usually limited, which means time-varying delays are usually a general form of time delays. Thus, it is necessary to take time-varying delays into account. For example, Liu *et al.* [12] considered the synchronization of CDNs and their linear coupling form is shown as $\Gamma(x_h(t - \tau_1(t)) - x_k(t - \tau_2(t)))$, where self-delay and neighboring delay are different. However, the internal delays and coupling delays were assumed to be the same in their work. As a matter of fact, internal delays should be different from coupling delays in reality. Therefore, it is more general and practical to study CDNs with internal time-varying delays and coupling time-varying delays that are composed of different self-delays and neighboring delays.

Besides, the coupling structures in most of the literature are set to be constant. In other words, the coupling structures of CDNs remain the same all the time. In [4], the coupling structure of CDNs was set to be constant. However, in real cases, the coupling structure of CDNs is time varying. For example, in biomathematics, the dispersal rate of species in different communities will change as time goes on [13]. Therefore, CDNs with time-varying coupling structure are more general in real world and have wider applications. To the best of our knowledge, few researches for CDNs with time-varying coupling structure have been done in recent published papers. In this paper, we will explore some effective synchronization criteria for CDNs with time-varying coupling structure.

Recently, many intermittent control strategies and approaches have been proposed for lots of purposes in engineering fields since intermittent control can reduce the amount of transmitted information and have wider applications [14]–[21]. For this reason, intermittent control becomes a hot topic in many researches. Huang *et al.* [22] used intermittent linear control to synchronize delayed chaotic

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systems. Xia and Cao [23] used periodically intermittent control to discuss the synchronization of CDNs. In the aforementioned researches, intermittent control is set to be periodical, which means the control period and control width are fixed. However, in real world, some applications, such as the generation of wind power [24], may be restricted because of the feature about constant control period and control width. Hence, a more general control scheme based on changeable control period and control width called aperiodically intermittent control (AIC) has been successfully developed to synchronize CDNs [25]–[28]. Liu and Chen [25] proposed pinning AIC to achieve synchronization of nonlinear coupled networks. In [26], aperiodically intermittent pinning control laws were discussed for the quasi-synchronization problem of nonlinear coupled chaotic systems. In addition, authors used aperiodically adaptive intermittent pinning control to study the outer synchronization between two hybrid-coupled delayed dynamical networks in [27]. In [28], delay-independent stabilization of a class of time-delay systems via periodically intermittent control was considered. Other related works can also be found in [29]–[34]. However, these works treat coupling structure as constant, or time-varying delays are not considered into their models. Thus, in this paper, we will study the synchronization problem of time-varying CDNs with multiple time-varying delays (TCDNMTDs) based on AIC.

Motivated by the aforementioned discussion, in this paper, we mainly investigate the p th moment exponential synchronization (PMES) of TCDNMTDs via AIC. By using the graph-theoretic approach and Lyapunov method, a global Lyapunov function is established and some effective synchronization criteria for TCDNMTDs are shown. It is worthwhile to notice that the coupling structure of our CDNs is time varying. Also, internal delays, self-delay, and neighboring delay are included, and they are different from each other. Moreover, we apply theoretical results to second-order Kuramoto oscillators (SKOs) and coupled Chua's circuits and some useful criteria for their exponential synchronization are further obtained.

The outline of this paper is as follows. In Section II, some useful preliminaries are proposed. The main results in this paper are given in Section III. In Sections IV and V, our theoretical results are applied to SKOs and coupled Chua's circuits, then two numerical simulations are presented.

II. PRELIMINARIES AND MODEL FORMULATIONS

Let $\mathcal{N} = \{1, 2, \dots, N\}$, $\mathbb{L} = \{1, 2, \dots, n\}$, $\mathbb{R}_{\geq 0} = [0, +\infty)$, and $\mathbb{Z}^+ = \{1, 2, \dots\}$. $\mathbb{R}^n$ denotes the n -dimensional Euclidean space. If $x \in \mathbb{R}^n$, then $|x|$ is Euclidean norm. A complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfies the usual conditions. Meanwhile, $\mathbb{E}(\cdot)$ stands for the mathematical expectation with respect to the given probability measure $\mathbb{P}$. And $B(t)$ is a one-dimensional (1-D) Brownian motion defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. And Dini derivative of function f is denoted by D^+f .

Now, let $\mathcal{G} = (v, \varepsilon, A(t))$ be a weighted digraph of order with the set of nodes $v = \{v_1, v_2, \dots, v_N\}$, set of edges

$\varepsilon \subseteq v \times v$, and a weighted adjacency matrix $A(t) = (a_{kh}(t))_{N \times N}$ with adjacency elements $a_{kh}(t) \geq 0$. The adjacency elements associated with the edges of the digraph are positive. Meanwhile, we assume $a_{kk}(t) = 0$ for $k \in \mathcal{N}$. A digraph is called strongly connected if and only if any two distinct nodes of the digraph can be connected via a path that follows the direction of the edges of the digraph. The Laplacian matrix of digraph $(\mathcal{G}, A(t))$ is given by $L(t) = (l_{kh}(t))_{N \times N}$, where $l_{kh}(t) = -a_{kh}(t)$, for $k \neq h$, $l_{kk}(t) = \sum_{h \neq k} a_{kh}(t)$.

Next, we consider general TCDNMTDs with N nodes in the following form:

$$\begin{aligned} dx_k(t) = & \left(f_k(x_k(t), x_k(t - \tau_1(t)), t) \right. \\ & + \sum_{h=1}^N a_{kh}(t) H_{kh}(x_k(t - \tau_2(t)), x_h(t - \tau_3(t))) \\ & \left. + g_k(x_k(t - \tau_4(t)), t) dB(t), k \in \mathcal{N} \right) dt \end{aligned} \quad (1)$$

where $x_k \in \mathbb{R}^n$ is the state vector, functions $f_k : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ and $g_k : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ are nonlinear vector-valued continuous functions, and $H_{kh} : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ stands for the influence function from node h to node k . And $a_{kh}(t)$ denotes the time-varying coupling strength. Time-varying delays $\tau_i(t)$ ($i = 1, 2, 3, 4$) are bounded, that is, $0 \leq \tau_i(t) \leq \tau$. $\tau_1(t)$ and $\tau_4(t)$ are internal delays, $\tau_2(t)$ denotes the self-delay, and $\tau_3(t)$ stands for the neighboring delay.

Remark 1: In fact, time-varying delays play an important part in real life and many interesting results have been reported [18], [35], [36]. In these works, the range of derivatives of time-varying delays is restricted, which is $\dot{\tau}(t) < c < 1$ ($c > 0$). However, in this paper, the restriction condition of time-varying delays has been removed. Moreover, internal time-varying delays and coupling time-varying delays are taken into account and they are different from each other, while many works [37]–[39] regarded internal and coupling time-varying delays as the same. Therefore, our results are more general.

Our main purpose is to utilize AIC to make two general TCDNMTDs achieve PMES. Hence, regarding system (1) as a drive system, the response system with AIC is given by

$$\begin{aligned} dy_k(t) = & \left(f_k(y_k(t), y_k(t - \tau_1(t)), t) + \sum_{h=1}^N a_{kh}(t) \right. \\ & \times H_{kh}(y_k(t - \tau_2(t)), y_h(t - \tau_3(t))) + u_k(t) \left. \right) dt \\ & + g_k(y_k(t - \tau_4(t)), t) dB(t), k \in \mathcal{N} \end{aligned} \quad (2)$$

where $y_k \in \mathbb{R}^n$ denotes the state vector of the k th subsystem and

$$u_k(t) = \Upsilon_k(t)(y_k(t) - x_k(t)) \quad (3)$$

in which

$$\Upsilon_k(t) = \begin{cases} -\Upsilon_k, & t_m \leq t < s_m \\ 0, & s_m \leq t < t_{m+1} \end{cases} \quad m = 0, 1, \dots$$

where $\Upsilon_k > 0$ stands for the intermittent control strength, and $[t_m, t_{m+1})$ is a control period (see [25] for details). We define

$\Psi = \limsup_{m \rightarrow \infty} [(t_{m+1} - s_m)/(t_{m+1} - t_m)]$. Obviously, $0 \leq \Psi < 1$. When $s_m - t_m = T_1$ and $t_{m+1} - s_m = T_2$, AIC becomes periodically intermittent control. Particularly, intermittent control can become continuous control when $T_2 = 0$.

We define $e_k(t) = y_k(t) - x_k(t)$ as the synchronization error vector. Then we obtain the following error system:

$$\begin{aligned} de_k(t) = & \left(\tilde{f}_k(e_k(t), e_k(t - \tau_1(t)), t) + \sum_{h=1}^N a_{kh}(t) \right. \\ & \times \tilde{H}_{kh}(e_k(t - \tau_2(t)), e_h(t - \tau_3(t))) + u_k(t) \Big) dt \\ & + \tilde{g}_k(e_k(t - \tau_4(t)), t) dB(t), \quad k \in \mathcal{N} \end{aligned} \quad (4)$$

where $\tilde{f}_k(e_k(t), e_k(t - \tau_1(t)), t) = f_k(y_k(t), y_k(t - \tau_1(t)), t) - f_k(x_k(t), x_k(t - \tau_1(t)), t)$, $\tilde{g}_k(e_k(t - \tau_4(t)), t) = g_k(y_k(t - \tau_4(t)), t) - g_k(x_k(t - \tau_4(t)), t)$, $\tilde{H}_{kh}(e_k(t - \tau_2(t)), e_h(t - \tau_3(t))) = H_{kh}(y_k(t - \tau_2(t)), y_h(t - \tau_3(t))) - H_{kh}(x_k(t - \tau_2(t)), x_h(t - \tau_3(t)))$.

Subsequently, some basic definitions are given.

Definition 1: Write $C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0}; \mathbb{R}_{\geq 0})$ for the family of all non-negative functions $V(x, t)$ on $\mathbb{R}^n \times \mathbb{R}_{\geq 0}$ that are continuously twice differentiable in x and once in t . Suppose $V_k(e_k, t) \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0}; \mathbb{R}_{\geq 0})$, then differential operator $\mathcal{L}V_k(e_k, t)$ is defined as

$$\begin{aligned} \mathcal{L}V_k(e_k, t) = & \frac{\partial V_k(e_k, t)}{\partial t} + \frac{\partial V_k(e_k, t)}{\partial e_k} \\ & \times \left[\tilde{f}_k(e_k(t), e_k(t - \tau_1(t)), t) + \sum_{h=1}^N a_{kh}(t) \tilde{H}_{kh} \right. \\ & \times (e_k(t - \tau_2(t)), e_h(t - \tau_3(t))) + u_k(t) \Big] \\ & + \frac{1}{2} \text{trace} \left[\left[\tilde{g}_k(e_k(t - \tau_4(t)), t) \right]^T \right. \\ & \times \left. \frac{\partial^2 V_k(e_k, t)}{\partial e_k^2} \tilde{g}_k(e_k(t - \tau_4(t)), t) \right] \end{aligned}$$

where

$$\begin{aligned} \frac{\partial V_k(e_k, t)}{\partial e_k} = & \left(\frac{\partial V_k(e_k, t)}{e_k^{(1)}}, \frac{\partial V_k(e_k, t)}{e_k^{(2)}}, \dots, \frac{\partial V_k(e_k, t)}{e_k^{(n)}} \right) \\ \frac{\partial^2 V_k(e_k, t)}{\partial e_k^2} = & \left(\frac{\partial^2 V_k(e_k, t)}{\partial e_k^{(p)} \partial e_k^{(q)}} \right)_{n \times n}. \end{aligned}$$

Definition 2: Drive-response systems (1) and (2) are said to be PMES if

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log(\mathbb{E}|e(t)|^p) < 0$$

in which $e(t) = (e_1^T(t), e_2^T(t), \dots, e_N^T(t))^T \in \mathbb{R}^{nN}$. In particular, when $p = 2$, it is said to be exponential synchronization in mean square (ESMS).

Definition 3: There exists a function $V_k(e_k, t) \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0}; \mathbb{R}_{\geq 0})$. If the following conditions are satisfied.

(A₁): There are constants $r_k^{(1)} > 0$, $r_k^{(2)} > 0$, $\alpha_k^{(1)} > 0$, and $\alpha_k^{(2)} > 0$, and functions $F_{kh}, \delta_{kh}(t) \geq 0$, it follows:

$$\mathcal{L}V_k(e_k(t), t) \leq \begin{cases} -r_k^{(1)}|e_k(t)|^p + \alpha_k^{(1)} \sum_{i=1}^4 |e_k(t - \tau_i(t))|^p \\ \quad + \sum_{h=1}^N \delta_{kh}(t) F_{kh}, & t_m \leq t < s_m \\ r_k^{(2)}|e_k(t)|^p + \alpha_k^{(2)} \sum_{i=1}^4 |e_k(t - \tau_i(t))|^p \\ \quad + \sum_{h=1}^N \delta_{kh}(t) F_{kh}, & s_m \leq t < t_{m+1} \end{cases}$$

where $F_{kh} = F_{kh}(e_k(t - \tau_3(t)), e_h(t - \tau_3(t)))$.

(A₂): There exist positive constants $\beta_k^{(1)}, \beta_k^{(2)}$, and $p \geq 2$ such that

$$\beta_k^{(1)}|e_k|^p \leq V_k(e_k, t) \leq \beta_k^{(2)}|e_k|^p.$$

(A₃): Along each directed cycle C_Q of digraph $(\mathcal{G}, (\delta_{kh}(t))_{N \times N})$, it holds that

$$\sum_{(h,k) \in E(C_Q)} F_{kh}(e_k, e_h) \leq 0.$$

Then we call $V_k(e_k, t)$ as a vertex Lyapunov function.

III. MAIN RESULTS

In this section, the PMES of systems (1) and (2) is considered. The main results are given as follows.

For convenience, let $\epsilon_1 = \min_{k \in \mathcal{N}} \{r_k^{(1)}/\beta_k^{(2)}\}$, $\bar{\epsilon}_1 = \max_{k \in \mathcal{N}} \{\alpha_k^{(1)}/\beta_k^{(1)}\}$, $\epsilon_2 = \max_{k \in \mathcal{N}} \{r_k^{(2)}/\beta_k^{(1)}\}$, $\bar{\epsilon}_2 = \max_{k \in \mathcal{N}} \{\alpha_k^{(2)}/\beta_k^{(1)}\}$, $\epsilon^* = 4 \max\{\bar{\epsilon}_1, \bar{\epsilon}_2\}$, and $\rho = \epsilon_1 + \epsilon_2$.

Theorem 1: There exist vertex Lyapunov functions $V_k(e_k, t) \in C^{2,1}(\mathbb{R}^n \times \mathbb{R}_{\geq 0}; \mathbb{R}_{\geq 0})$ and digraph $(\mathcal{G}, (\delta_{kh}(t))_{N \times N})$, where $\delta'_{kh}(t) \leq 0$, is strongly connected. Systems (1) and (2) can reach PMES if the following conditions are established:

$$\epsilon_1 - \epsilon^* > 0, \quad \lambda - \rho\Psi > 0$$

where λ is the unique positive solution of the equation $\lambda - \epsilon_1 + \epsilon^*e^{\lambda\tau} = 0$.

Proof: We define function $g(\lambda) = \lambda - \epsilon_1 + \epsilon^*e^{\lambda\tau}$. According to condition (A₄), it is clear that $g(0) = -\epsilon_1 + \epsilon^* < 0$, $g(+\infty) > 0$ and $g'(\lambda) > 0$. Based on the monotonicity and continuity of $g(\lambda)$, the equation $g(\lambda) = 0$ has a unique positive solution λ . Then, we consider the following global Lyapunov function $V(e, t) = \sum_{k=1}^N d_k(t) V_k(e_k, t)$, where $d_k(t)$ denotes the cofactor of the k th diagonal element of the Laplacian matrix of digraph $(\mathcal{G}, \delta_{kh}(t)_{N \times N})$.

According to [40], we can obtain $d_k(t) = \sum_{\mathcal{T} \in \mathbb{T}_k} w(\mathcal{T})$, in which the spanning tree $\mathcal{T}$ composed of the product of time-varying weights $\delta_{kh}(t)$, and $w(\mathcal{T})$ is the weight of the spanning tree $\mathcal{T}$. Furthermore, since digraph $(\mathcal{G}, \delta_{kh}(t)_{N \times N})$ is strongly connected, it is easy to get that $d_k(t) > 0$. Therefore, based on $\delta'_{kh}(t) \leq 0$, we can derive easily that $d'_k(t) \leq 0$. Hence, $d_k(t)$ is bounded and there exist positive constants c_1 and c_2 satisfying $c_1 \leq d_k(t) \leq c_2$.

Moreover, according to condition (A₂), we acquire that

$$V(e, t) \geq \sum_{k=1}^N d_k(t) \beta_k^{(1)} |e_k|^p \geq \hat{\alpha} |e|^p \quad (5)$$

where $\hat{\alpha} = (c_1 \sum_{k=1}^N \beta_k^{(1)})^{1-(p/2)} (\min_{k \in \mathcal{N}} \{c_1 \beta_k^{(1)}\})^{(p/2)} > 0$. Meanwhile, we can also obtain the following result similarly:

$$V(e, t) \leq \sum_{k=1}^N d_k(t) \beta_k^{(2)} |e_k|^p \leq \beta |e|^p$$

where $\beta = (\sum_{k=1}^N c_2 \beta_k^{(2)}) > 0$.

Denote $Q(t) = W(t) - hM_0$, in which $h > 1$, $W(t) = e^{\lambda t} V(e(t), t)$, and $M_0 = \sup_{-\tau \leq s \leq 0} \{\beta |e(s)|^p\}$.

For $t \in [-\tau, 0]$, we can easily get

$$\mathbb{E}Q(t) < 0. \quad (6)$$

Next, for all $t \in [0, s_0]$, we prove that $\mathbb{E}Q(t) < 0$. Using the proof by contradiction, we assume that there exists $a_0 \in [0, s_0]$, such that for $t \in [0, a_0]$, one has

$$\mathbb{E}Q(a_0) = 0, D^+ \mathbb{E}Q(a_0) \geq 0, \mathbb{E}Q(t) < 0.$$

According to [41] and condition (A₁), integrating both sides of condition (A₁) from t to $t+h$ with $h > 0$, and taking the mathematical expectation, one has

$$\frac{\mathbb{E}Q(t+h) - \mathbb{E}Q(t)}{h} = \frac{1}{h} \int_t^{t+h} \mathbb{E} \mathcal{L}Q(s) ds.$$

Letting $h \rightarrow 0^+$, it can be deduced that

$$\begin{aligned} D^+ \mathbb{E}Q(a_0) &= \mathbb{E} \mathcal{L}Q(a_0) \\ &= \mathbb{E}(\lambda e^{\lambda a_0} V(e(a_0), a_0) + e^{\lambda a_0} \mathcal{L}V(e(a_0), a_0)) \\ &\leq e^{\lambda a_0} \left[\lambda \mathbb{E}V(e(a_0), a_0) - \sum_{k=1}^N d_k(a_0) r_k^{(1)} \mathbb{E}|e_k(a_0)|^p \right. \\ &\quad + \mathbb{E} \sum_{k=1}^N \sum_{i=1}^4 d_k(a_0) \alpha_k^{(1)} |e_k(a_0 - \tau_i(a_0))|^p \\ &\quad \left. + \mathbb{E} \sum_{k=1}^N \sum_{h=1}^N d_k(a_0) \delta_{kh}(a_0) F_{kh} \right] \\ &\leq \lambda e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) - \epsilon_1 e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) \\ &\quad + e^{\lambda a_0} \bar{\epsilon}_1 \mathbb{E} \sum_{i=1}^4 V(e(a_0 - \tau_i(a_0)), a_0 - \tau_i(a_0)) \\ &\quad + e^{\lambda a_0} \mathbb{E} \sum_{k=1}^N \sum_{h=1}^N d_k(a_0) \delta_{kh}(a_0) F_{kh}. \end{aligned} \quad (7)$$

According to [40] and condition (A₃), one has

$$\sum_{k=1}^N \sum_{h=1}^N d_k(a_0) \delta_{kh}(a_0) F_{kh} = \sum_{Q \in \mathcal{Q}} w(Q) \sum_{(s,r) \in E(C_Q)} F_{rs} \leq 0.$$

Combining (7) with (8), it is obvious that

$$\begin{aligned} D^+ \mathbb{E}Q(a_0) &\leq \lambda e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) - \epsilon_1 e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) \\ &\quad + 4e^{\lambda a_0} \bar{\epsilon}_1 \mathbb{E} \sup_{a_0 - \tau \leq s \leq a_0} \{V(e(s), s)\}. \end{aligned}$$

Meanwhile, one can see that

$$e^{\lambda a_0} \mathbb{E} \sup_{a_0 - \tau \leq s \leq a_0} \{V(e(s), s)\} < e^{\lambda \tau} \mathbb{E}W(a_0).$$

Therefore, we derive that

$$\begin{aligned} D^+ \mathbb{E}Q(a_0) &< \lambda e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) - \epsilon_1 e^{\lambda a_0} \mathbb{E}V(e(a_0), a_0) \\ &\quad + 4\bar{\epsilon}_1 e^{\lambda \tau} \mathbb{E}W(a_0) \\ &< (\lambda - \epsilon_1 + \epsilon^* e^{\lambda \tau}) \mathbb{E}W(a_0) = 0 \end{aligned}$$

which contradicts the inequality $D^+ \mathbb{E}Q(a_0) \geq 0$ and implies that $\mathbb{E}Q(t) < 0$ holds. Moreover, we can obtain

$$\mathbb{E}H(t) \triangleq \mathbb{E}(W(t) - hM_0 e^{\rho(t-s_0)}) < 0, \text{ for all } t \in [s_0, t_1].$$

Otherwise, there is $a_1 \in [s_0, t_1]$ such that for $t \in [s_0, a_1]$, one has

$$\mathbb{E}H(a_1) = 0, D^+ \mathbb{E}H(a_1) \geq 0, \mathbb{E}H(t) < 0. \quad (9)$$

According to condition (A₁), one has

$$\begin{aligned} D^+ \mathbb{E}H(a_1) &\leq e^{\lambda a_1} \left[\lambda \mathbb{E}V(e(a_1), a_1) + \sum_{k=1}^N d_k(a_1) \right. \\ &\quad \times \mathbb{E} \left(r_k^{(2)} |e_k(a_1)|^p + \alpha_k^{(2)} \sum_{i=1}^4 |e_k(a_1 - \tau_i(a_1))|^p \right. \\ &\quad \left. \left. + \sum_{h=1}^N \delta_{kh}(a_1) F_{kh} \right) \right] - h \mathbb{E}M_0 \rho e^{\rho(a_1-s_0)} \\ &\leq e^{\lambda a_1} \left[\lambda \mathbb{E}V(e(a_1), a_1) + \sum_{k=1}^N d_k(a_1) r_k^{(2)} \mathbb{E}|e_k(a_1)|^p \right. \\ &\quad + \sum_{i=1}^4 \sum_{k=1}^N d_k(a_1) \frac{\alpha_k^{(2)}}{\beta_k^{(1)}} \mathbb{E}V_k(e_k(a_1 - \tau_i(a_1)) \\ &\quad \times a_1 - \tau_i(a_1)) \\ &\quad \left. + \mathbb{E} \sum_{k=1}^N \sum_{h=1}^N d_k(a_1) \delta_{kh}(a_1) F_{kh} \right] \\ &\quad - h \mathbb{E}M_0 \rho e^{\rho(a_1-s_0)}. \end{aligned}$$

According to conditions (A₂), (A₃), (A₄), and (8), one get

$$\begin{aligned} D^+ \mathbb{E}H(a_1) &\leq \lambda e^{\lambda a_1} \mathbb{E}V(e(a_1), a_1) + \epsilon_2 e^{\lambda a_1} \mathbb{E}V(e(a_1), a_1) \\ &\quad + 4\bar{\epsilon}_2 e^{\lambda a_1} \mathbb{E} \sup_{a_1 - \tau \leq s \leq a_1} \{V(e(s), s)\} \\ &\quad - h \mathbb{E}M_0 \rho e^{\rho(a_1-s_0)}. \end{aligned} \quad (10)$$

In fact, if $s_0 < a_1 - \tau < a_1$, then

$$\begin{aligned} \sup_{a_1 - \tau \leq s \leq a_1} \{V(e(s), s)\} &< e^{-\lambda(a_1 - \tau)} h \mathbb{E}M_0 e^{\rho(a_1-s_0)} \\ &= e^{\lambda \tau} V(e(a_1), a_1) \end{aligned} \quad (11)$$

and if $-\tau < a_1 - \tau < s_0$, we can obtain

$$\begin{aligned} \sup_{a_1 - \tau \leq s \leq a_1} \{V(e(s), s)\} &= \max \left\{ \sup_{a_1 - \tau \leq s \leq s_0} \{V(e(s), s)\}, \sup_{s_0 \leq s \leq a_1} \{V(e(s), s)\} \right\} \\ &\leq \max \left\{ h \mathbb{E}M_0 e^{-\lambda(a_1 - \tau)}, e^{\lambda(a_1 - s_0)} V(e(a_1), a_1) \right\} \\ &\quad \times e^{-\lambda(a_1 - \tau)} h \mathbb{E}M_0 e^{\rho(a_1-s_0)} \\ &\leq e^{\lambda \tau} V(e(a_1), a_1). \end{aligned} \quad (12)$$

Therefore, combining (11) with (12), it shows

$$\sup_{a_1 - \tau \leq s \leq a_1} \{V(e(s), s)\} < e^{\lambda \tau} V(e(a_1), a_1). \quad (13)$$

Using (10), (13), and definition of $W(t)$, one has

$$\begin{aligned} D^+ \mathbb{E}H(a_1) &< (\lambda + \epsilon_2 - \rho) \mathbb{E}W(a_1) + 4\bar{\epsilon}_2 e^{\lambda a_1} \\ &\quad \times \mathbb{E} \sup_{a_1 - \tau \leq s \leq a_1} \{V(e(s), s)\} \\ &< (\lambda - \epsilon_1 + \epsilon^* e^{\lambda \tau}) \mathbb{E}W(a_1) = 0 \end{aligned}$$

which contradicts the inequality (9). Therefore, $\mathbb{E}H(t) < 0$, for all $t \in [s_0, t_1)$, that is

$$\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho(t-s_0)} < h \mathbb{E}M_0 e^{\rho(t_1-s_0)}. \quad (14)$$

By virtue of (6) and (14), it is easy to get that for all $t \in [-\tau, t_1)$, one has $\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho(t_1-s_0)}$.

Moreover, for all $t \in [t_1, s_1)$, it follows $\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho(t_1-s_0)}$, and for all $t \in [s_1, t_2)$, it has $\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho[(t_1-s_0)+(t-s_1)]}$. With mathematical induction, we can get that for $t \in [t_m, s_m)$, one has

$$\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho \sum_{k=1}^m (t_k - s_{k-1})}$$

and for $t \in [s_m, t_{m+1})$, one has

$$\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho \left[\sum_{k=1}^m (t_k - s_{k-1}) + (t - s_m) \right]}.$$

For all $t \in \mathbb{R}_{\geq 0}$, there is $m \in \mathbb{Z}^+$ such that $t \in [t_m, t_{m+1})$. When $t \in [t_m, s_m)$, one has

$$\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho \sum_{k=1}^m (t_k - s_{k-1})} \leq h \mathbb{E}M_0 e^{\rho \Psi t}.$$

For all $t \in [s_m, t_{m+1})$, we have the following inequality similarly:

$$\mathbb{E}W(t) < h \mathbb{E}M_0 e^{\rho \left[\sum_{k=1}^m (t_k - s_{k-1}) + (t - s_m) \right]} \leq h \mathbb{E}M_0 e^{\rho \Psi t}.$$

Letting $h \rightarrow 1$ and according to $W(t)$, we get

$$\mathbb{E}V(e(t), t) \leq \mathbb{E}M_0 e^{-(\lambda - \rho \Psi)t}. \quad (15)$$

According to (5) and (15), we acquire that

$$\mathbb{E}|e(t)|^p \leq \frac{1}{\alpha} \mathbb{E}M_0 e^{-(\lambda - \rho \Psi)t}. \quad (16)$$

Therefore, according to Definition 2, we can draw the conclusion that the PMES of systems (1) and (2) is realized. ■

Remark 2: Based on the graph theory and Lyapunov method, a criterion for the PMES of systems (1) and (2) is established in Theorem 1. Also, we propose the global Lyapunov function $V(e, t) = \sum_{k=1}^N d_k(t) V_k(e_k, t)$, where $d_k(t)$ is defined in the proof of Theorem 1, which shows that synchronization is closely related to the topology structure of digraph $(\mathcal{G}, (\delta_{kh}(t))_{N \times N})$.

Remark 3: Recently, a lot of researchers have used linear matrix inequalities (LMIs) to study the dynamical properties of CDNs [45], [46]. The coupling structure of CDNs is required to be constant since a_{kh} appears in LMIs.

However, in this paper, by combining (8) in Theorem 1 with graph theory [40], we provide an important inequality $\sum_{k=1}^N \sum_{h=1}^N d_k(t) \delta_{kh}(t) F_{kh} \leq 0$. This indicates $\delta_{kh}(t)$ has a positive effect on PMES.

Theorem 2: For any $k, h \in \mathcal{N}$, systems (1) and (2) can achieve PMES if the following conditions are satisfied.

(B₁): The following inequalities hold:

$$\begin{aligned} |\tilde{f}_k(e_k, \tilde{e}_k, t)| &\leq a_k |e_k| + b_k |\tilde{e}_k|, |\tilde{g}_k(\tilde{e}_k, t)|^2 \leq \delta_k |\tilde{e}_k|^2 \\ |\tilde{H}_{kh}(e_k, e_h)| &\leq \epsilon_{kh} (|e_k| + |e_h|) \end{aligned}$$

where $\delta_k, b_k, \epsilon_{kh}$, and a_k are positive constants.

(B₂): There exists a positive constant $p \geq 2$ such that

$$\begin{aligned} -\chi_k &\triangleq p a_k + (p-1) b_k + (p-1)(p-2) \delta_k - p \Upsilon_k \\ &\quad + (2p-1) \sum_{h=1}^N a_{kh} \epsilon_{kh} \leq 0. \end{aligned}$$

(B₃): Assume that $a'_{kh}(t) \leq 0$ and digraph $(\mathcal{G}, A(t))$ is strongly connected, in which $A(t) = (a_{kh}(t))_{N \times N}$.

(B₄): Assume that $\epsilon_1 = \min_{k \in \mathcal{N}} \{\chi_k / \beta_k^{(2)}\}$, $\bar{\epsilon}_1 = \min_{k \in \mathcal{N}} \{\eta_k / \beta_k^{(1)}\}$, $\epsilon_2 = \min_{k \in \mathcal{N}} \{p \Upsilon_k - \chi_k / \beta_k^{(1)}\}$, and $\bar{\epsilon}_2 = \max_{k \in \mathcal{N}} \{\eta_k / \beta_k^{(1)}\}$, where $\eta_k = \max\{b_k, 2(p-1)\delta_k, \sum_{h=1}^N a_{kh} \epsilon_{kh}\}$. $\epsilon^* = \max\{4\bar{\epsilon}_1, 4\bar{\epsilon}_2\}$, and $\epsilon_1 - \epsilon^* > 0$. Moreover, $\lambda - \rho \Psi > 0$, where $\rho = \epsilon_1 + \epsilon_2$ and λ is the unique positive solution of $\lambda - \epsilon_1 + \epsilon^* e^{\lambda \tau} = 0$.

Proof: First, set function $V_k(e_k, t) = |e_k|^p$. For any $k \in \mathcal{N}$, it is clear that condition (A₂) is satisfied. When $t \in [t_m, s_m)$, based on Definition 1 and condition (B₁), one has

$$\begin{aligned} \mathcal{L}V_k(e_k(t), t) &\leq p |e_k(t)|^{p-2} |e_k(t)| [a_k |e_k(t)| + b_k |e_k(t - \tau_1(t))|] \\ &\quad + p(p-1) \delta_k |e_k(t)|^{p-2} |e_k(t - \tau_4(t))|^2 \\ &\quad - p \Upsilon_k |e_k(t)|^p + p |e_k(t)|^{p-1} \sum_{h=1}^N a_{kh}(t) \\ &\quad \times |\tilde{H}_{kh}(e_k(t - \tau_2(t)), e_h(t - \tau_3(t)))|. \quad (17) \end{aligned}$$

Based on the Young's inequality

$$|\theta|^\zeta |\vartheta|^\varsigma \leq \frac{\zeta}{\zeta + \varsigma} |\theta|^{\zeta + \varsigma} + \frac{\varsigma}{\zeta + \varsigma} |\vartheta|^{\zeta + \varsigma}$$

we can get that

$$\begin{aligned} &p |e_k(t)|^{p-1} \sum_{h=1}^N a_{kh}(t) |\tilde{H}_{kh}(e_k(t - \tau_2(t)), e_h(t - \tau_3(t)))| \\ &\leq p |e_k(t)|^{p-1} \sum_{h=1}^N a_{kh}(t) \epsilon_{kh} (|e_k(t - \tau_2(t))| + |e_h(t - \tau_3(t))|) \\ &\leq (2p-1) \sum_{h=1}^N a_{kh}(t) \epsilon_{kh} |e_k(t)|^p + \sum_{h=1}^N a_{kh}(t) \epsilon_{kh} |e_k(t - \tau_2(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t) \epsilon_{kh} |e_k(t - \tau_3(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t) \epsilon_{kh} (|e_h(t - \tau_3(t))|^p - |e_k(t - \tau_3(t))|^p) \quad (18) \end{aligned}$$

and we can get the following inequalities:

$$\begin{aligned} |e_k(t)|^{p-2}|e_k(t-\tau_4(t))|^2 &\leq \frac{p-2}{p}|e_k(t)|^p + \frac{2}{p}|e_k(t-\tau_4(t))|^p \\ |e_k(t)|^{p-1}|e_k(t-\tau_1(t))| &\leq \frac{p-1}{p}|e_k(t)|^p + \frac{1}{p}|e_k(t-\tau_1(t))|^p. \end{aligned} \quad (19)$$

Combining (17) and (18) with (19), we can derive that

$$\begin{aligned} \mathcal{L}V_k(e_k(t), t) &\leq pa_k|e_k(t)|^p + b_k((p-1)|e_k(t)|^p + |e_k(t-\tau_1(t))|^p) \\ &\quad + (p-1)\delta_k((p-2)|e_k(t)|^p + 2|e_k(t-\tau_4(t))|^p) \\ &\quad - p\Upsilon_k|e_k(t)|^p + (2p-1)\sum_{h=1}^N a_{kh}(t)\epsilon_{kh}|e_k(t)|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t)\epsilon_{kh}|e_k(t-\tau_2(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t)\epsilon_{kh}|e_k(t-\tau_3(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t)\epsilon_{kh}(|e_h(t-\tau_3(t))|^p - |e_k(t-\tau_3(t))|^p) \\ &\leq -\chi_k|e_k(t)|^p + \eta_k \sum_{i=1}^4 |e_k(t-\tau_i(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t)\epsilon_{kh}(|e_h(t-\tau_3(t))|^p - |e_k(t-\tau_3(t))|^p). \end{aligned} \quad (20)$$

For $t \in [s_m, t_{m+1})$, similarly, we can obtain that

$$\begin{aligned} \mathcal{L}V_k(e_k(t), t) &\leq (p\Upsilon_k - \chi_k)|e_k(t)|^p + \eta_k \sum_{i=1}^4 |e_k(t-\tau_i(t))|^p \\ &\quad + \sum_{h=1}^N a_{kh}(t)\epsilon_{kh}(|e_h(t-\tau_3(t))|^p - |e_k(t-\tau_3(t))|^p). \end{aligned}$$

Therefore, one has

$$\mathcal{L}V_k(e_k(t), t) \leq \begin{cases} -r_k^{(1)}|e_k(t)|^p + \alpha_k^{(1)} \sum_{i=1}^4 |e_k(t-\tau_i(t))|^p \\ \quad + \sum_{h=1}^N \delta_{kh}(t)F_{kh}, & t_m \leq t < s_m \\ r_k^{(2)}|e_k(t)|^p + \alpha_k^{(2)} \sum_{i=1}^4 |e_k(t-\tau_i(t))|^p \\ \quad + \sum_{h=1}^N \delta_{kh}(t)F_{kh}, & s_m \leq t < t_{m+1} \end{cases}$$

where $\delta_{kh}(t) = \epsilon_{kh}a_{kh}(t)$ and $F_{kh} = |e_h(t-\tau_3(t))|^p - |e_k(t-\tau_3(t))|^p$. Therefore, condition (A₁) is satisfied. Combining condition (B₃) with condition (B₄), all conditions of Theorem 1 are satisfied. Hence, drive-response systems (1) and (2) can achieve PMES. This completes the proof. ■

Remark 4: Recently, many works have been done to study the synchronization problem of CDNs [12], [42]–[44]. In these literatures, the coupling structures of CDNs were assumed to be constant. Different from previous work, the coupling structure considered in this paper is time varying. Moreover, according to condition (B₂) in Theorem 2, we know

$a'_{kh}(t) \leq 0$. Especially when $a'_{kh}(t) = 0$, which means the coupling structure of CDNs is constant, our conclusion is also available.

Remark 5: In Theorem 2, we require $p \geq 2$. As a matter of fact, based on the estimation properties of moment, our criterion is also established for $0 < p < 2$. Therefore, one corollary is stated as follows.

Corollary 1: All the conditions in Theorem 2 are satisfied. Then, p th ($0 < p < 2$) moment exponential synchronization of systems (1) and (2) can be realized.

Proof: From Theorem 2, it is easy to obtain that mean square exponential synchronization of systems (1) and (2) is achieved. According to Definition 2, one has

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log(\mathbb{E}|e(t)|^2) < 0.$$

Then, when $0 < p < 2$, based on the Hölder's inequality, we get that

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log(\mathbb{E}|e(t)|^p) \leq \frac{p}{2} \limsup_{t \rightarrow \infty} \frac{1}{t} \log(\mathbb{E}|e(t)|^2) < 0$$

which indicates that systems (1) and (2) are the p th moment exponentially synchronized. ■

IV. APPLICATION TO SKOS

As we all know, the synchronization analysis of SKOs is essential in power grid [47], [48]. Hence, it is very necessary to study the synchronization of SKOs. In [48], the SKOs are given as follows:

$$\ddot{x}_k = -\alpha\dot{x}_k + \Omega_k + \bar{\lambda} \sum_{h=1}^N a_{kh} \sin(x_h(t) - x_k(t)), \quad k \in \mathcal{N}$$

where α stands for the dissipating parameter, $\bar{\lambda}$ is the coupling strength, and Ω_k is the natural frequency of oscillators.

Next, the SKOs with stochastic interference and time-varying delays are considered in the following form:

$$\begin{aligned} \ddot{x}_k + \alpha\dot{x}_k - \Omega_k - \eta x_k(t - \tau_1(t)) \\ - \bar{\lambda} \sum_{h=1}^N a_{kh}(t) \sin(x_k(t - \tau_2(t)) - x_h(t - \tau_3(t))) \\ = rx_k(t - \tau_4(t))\dot{B}(t), \quad k \in \mathcal{N}. \end{aligned} \quad (21)$$

To research conveniently, let $\tilde{x}_k(t) = \dot{x}_k(t) + \xi x_k(t)$, where $\xi > 0$, then system (21) could be converted to

$$\begin{cases} d\tilde{x}_k(t) = (\tilde{x}_k(t) - \xi x_k(t))dt \\ d\tilde{x}_k(t) = [(\xi - \alpha)\tilde{x}_k(t) + (\xi\alpha - \xi^2)x_k(t) \\ \quad + \Omega_k + \eta x_k(t - \tau_1(t)) + \sum_{h=1}^N \bar{\lambda} a_{kh}(t) \\ \quad \sin(x_k(t - \tau_2(t)) - x_h(t - \tau_3(t)))]dt \\ \quad + rx_k(t - \tau_4(t))dB(t), \quad k \in \mathcal{N}. \end{cases} \quad (22)$$

In order to simplify writing, we define $X_k(t) = (x_k(t), \tilde{x}_k(t))^T$ and $F_k(X_k(t), X_k(t - \tau_1(t)), t) = (F_{k1}, F_{k2})^T$, where $F_{k1} = \tilde{x}_k(t) - \xi x_k(t)$, $F_{k2} = (\xi - \alpha)\tilde{x}_k(t) + (\xi\alpha - \xi^2)x_k(t) + \Omega_k + \eta x_k(t - \tau_1(t))$, $H_{kh}(X_k(t - \tau_2(t)), X_h(t - \tau_3(t))) = (0, \sin(x_k(t - \tau_2(t)) - x_h(t - \tau_3(t))))^T$, and $G_k(X_k(t - \tau_4(t)), t) = (0, rx_k(t - \tau_4(t)))^T$.

Hence, system (22) could be transformed to

$$\begin{aligned} dX_k(t) = & \left[F_k(X_k(t), X_k(t - \tau_1(t)), t) + \sum_{h=1}^N \bar{\lambda} a_{kh}(t) \right. \\ & \left. H_{kh}(X_k(t - \tau_2(t)), X_h(t - \tau_3(t))) \right] dt \\ & + G_k(X_k(t - \tau_4(t)), t) dB(t), \quad k \in \mathcal{N}. \end{aligned} \quad (23)$$

Regarding system (23) as a drive system, the response system with AIC is described as

$$\begin{aligned} dY_k(t) = & \left[F_k(Y_k(t), Y_k(t - \tau_1(t)), t) + \sum_{h=1}^N \bar{\lambda} a_{kh}(t) \right. \\ & \left. H_{kh}(Y_k(t - \tau_2(t)), Y_h(t - \tau_3(t))) + U_k(t) \right] dt \\ & + G_k(Y_k(t - \tau_4(t)), t) dB(t), \quad k \in \mathcal{N} \end{aligned} \quad (24)$$

where

$$U_k(t) = \Upsilon_k(t)(Y_k(t) - X_k(t)), \quad k \in \mathcal{N} \quad (25)$$

in which

$$\Upsilon_k(t) = \begin{cases} -\Upsilon_k, & t_m \leq t < s_m \\ 0, & s_m \leq t < t_{m+1} \end{cases} \quad m = 0, 1, \dots$$

here Υ_k is a positive constant. Let $E_k(t) = Y_k(t) - X_k(t)$ represent the synchronization error vector. According to systems (23) and (24), we can get the following:

$$\begin{aligned} dE_k(t) = & \left[\tilde{F}_k(E_k(t), E_k(t - \tau_1(t)), t) + \sum_{h=1}^N \bar{\lambda} a_{kh}(t) \right. \\ & \left. \times \tilde{H}_{kh}(E_k(t - \tau_2(t)), E_h(t - \tau_3(t))) + U_k(t) \right] dt \\ & + \tilde{G}_k(E_k(t - \tau_4(t)), t) dB(t), \quad k \in \mathcal{N} \end{aligned} \quad (26)$$

where $\tilde{F}_k(E_k(t), E_k(t - \tau_1(t)), t) = F_k(Y_k(t), Y_k(t - \tau_1(t)), t) - F_k(X_k(t), X_k(t - \tau_1(t)), t)$, $\tilde{G}_k(E_k(t - \tau_4(t)), t) = G_k(Y_k(t - \tau_4(t)), t) - G_k(X_k(t - \tau_4(t)), t)$, $\tilde{H}_{kh}(E_k(t - \tau_2(t)), E_h(t - \tau_3(t))) = H_{kh}(Y_k(t - \tau_2(t)), Y_h(t - \tau_3(t))) - H_{kh}(X_k(t - \tau_2(t)), X_h(t - \tau_3(t)))$.

Next, we will present a criterion about PMES of drive-response systems (23) and (24).

Theorem 3: For any $k, h \in \mathcal{N}$, $p \geq 2$, systems (23) and (24) can reach PMES if the following conditions hold.

(C₁): Assume that $a'_{kh}(t) \leq 0$ and digraph $(\mathcal{G}, (a_{kh}(t))_{N \times N})$ is strongly connected.

(C₂): Assume that $\epsilon_1 = \min_{k \in \mathcal{N}} \{\chi_k / \beta_k^{(2)}\}$, $\bar{\epsilon}_1 = \min_{k \in \mathcal{N}} \{\eta_k / \beta_k^{(1)}\}$, $\epsilon_2 = \min_{k \in \mathcal{N}} \{(p\Upsilon_k - \chi_k) / \beta_k^{(1)}\}$, $\bar{\epsilon}_2 = \max_{k \in \mathcal{N}} \{\eta_k / \beta_k^{(1)}\}$, and $\epsilon^* = \max\{4\bar{\epsilon}_1, 4\bar{\epsilon}_2\}$, where $\eta_k = \max\{b_k, 2(p-1)r^2, \sum_{h=1}^N a_{kh}(0)\sqrt{2\bar{\lambda}}\}$, $-\chi_k = pa_k + (p-1)b_k + (p-1)(p-2)r^2 - p\Upsilon_k + 2(p-1)\sum_{h=1}^N a_{kh}(0)\sqrt{2\bar{\lambda}} \leq 0$, where $a_k^2 = 2 + 3(\xi - \alpha)^2$, $b_k^2 = 2\xi^2 + 3(\xi\alpha - \xi^2)^2$. $\rho = \epsilon_1 + \epsilon_2$, such that $\epsilon_1 - \epsilon^* > 0$ and $\lambda - \rho\Psi > 0$, where λ is the unique positive solution of the equation $\lambda - \epsilon_1 + \epsilon^*e^{\lambda\tau} = 0$.

Proof: According to the definition of $\tilde{F}_k$, we derive

$$\begin{aligned} & |\tilde{F}_k(E_k(t), E_k(t - \tau_1(t)), t)|^2 \\ & \leq 2\tilde{e}_k^2(t) + 2\xi^2 e_k^2(t) + 3(\xi - \alpha)^2 \tilde{e}_k^2(t) \\ & \quad + 3(\xi\alpha - \xi^2)^2 e_k^2(t) + 3\eta^2 e_k^2(t - \tau_1(t)) \\ & \leq (a_k|E_k(t)| + b_k|E_k(t - \tau_1(t))|)^2. \end{aligned} \quad (27)$$

Hence, inequality (27) can be written as follows:

$$|\tilde{F}_k(E_k(t), E_k(t - \tau_1(t)), t)| \leq a_k|E_k(t)| + b_k|E_k(t - \tau_1(t))|. \quad (28)$$

Moreover

$$\begin{aligned} & |\tilde{G}_k(E_k(t - \tau_4(t)), t)| \leq r|E_k(t - \tau_4(t))| \\ & |\tilde{H}_{kh}(E_k(t - \tau_2(t)), E_h(t - \tau_3(t)))|^2 \\ & \leq \bar{\lambda}^2 |e_k(t - \tau_2(t)) - e_h(t - \tau_3(t))|^2 \\ & \leq 2\bar{\lambda}^2 (|E_k(t - \tau_2(t))| + |E_h(t - \tau_3(t))|)^2. \end{aligned} \quad (29)$$

Consequently

$$\begin{aligned} & |\tilde{H}_{kh}(E_k(t - \tau_2(t)), E_h(t - \tau_3(t)))| \\ & \leq \sqrt{2\bar{\lambda}} (|E_k(t - \tau_2(t))| + |E_h(t - \tau_3(t))|). \end{aligned} \quad (30)$$

By virtue of (28)–(30), we know that condition (B₁) is satisfied. Combining conditions (C₁) and (C₂) with (C₃), we know that conditions (B₂), (B₃), and (B₄) hold. Hence, according to Theorem 2, we can easily derive that PMES of systems (23) and (24) can be achieved. The proof is completed. ■

Remark 6: Many synchronization problems of SKOs have been widely investigated [4], [49], [50]. In [4], the outer synchronization problem of SKOs was considered. In [50], frequency synchronization of the SKOs was discussed. Different from previous work, by using the graph-theoretical method, the exponential synchronization problem for SKOs with stochastic interference and time-varying delays is considered.

Remark 7: Coupled oscillators have attracted a lot of attention recently [8], [39], [51], [52]. In [8], the exponential synchronization problem of coupled oscillators was investigated by Guo *et al.* In [51], the stabilization of stochastic coupled oscillators without time delay was studied. In this paper, we consider stochastic coupled oscillators with time-varying delays, which include internal delays, self-delays, and neighboring delays, and they are different from each other. Besides, many control schemes have been developed to investigate the dynamical properties of coupled oscillators. For example, Su *et al.* [53] studied exponential stability of complex networks via feedback control. Different from their researches, the control we use is AIC. Therefore, our theoretical results in this paper are more general.

Next, we will present some numerical simulations to demonstrate the practical applicability of our theoretical results. The following parameters related to system (22) with 100 nodes are given. Let $p = 2$, $r = 0.1$, $\xi = 0.05$, $\alpha = 0.5$, $\bar{\lambda} = 0.01$, $\eta = 0.05$, $\Omega_k = 0.001$, $\tau_1(t) = [0.04e^t / (1 + e^t)]$, $\tau_2(t) = [0.03e^{2t} / (1 + e^{2t})]$, $\tau_3(t) = [0.01e^t / (1 + e^t)]$, and

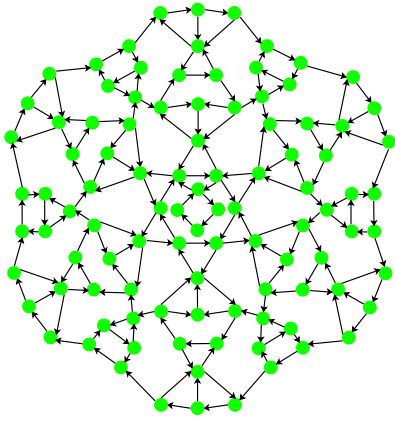


Fig. 1. Coupling structure of 100 coupled oscillators.

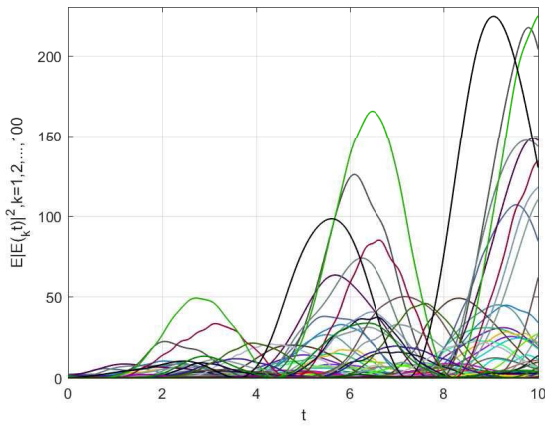


Fig. 2. Mean square figure of error system (26) without control.

$\tau_4(t) = [0.02e^{5t}/(1 + e^{5t})]$, $t \in [-0.04, 0]$. Moreover, the elements of weighted matrix $A(t) = (a_{kh}(t))_{100 \times 100}$ are given in Appendix A. Obviously, it holds that $a'_{kh}(t) \leq 0$ and $(\mathcal{G}, (a_{kh}(t))_{100 \times 100})$ is strongly connected (see Fig. 1).

From Fig. 2, it is easy to get that mean square exponential synchronization of systems (23) and (24) is not achieved. Therefore, it is necessary to design AIC such that systems (23) and (24) achieve exponential synchronization.

For simplicity, let $\Upsilon_1 = \Upsilon_2 = \dots = \Upsilon_{100} = 20.1$. We can get $a_k = 1.6148$ and $b_k = 0.0806$. Based on condition (C₃), it is easy to calculate that $\epsilon_1 = 26.5369$, $\bar{\epsilon}_1 = 0.0807$, $\epsilon_2 = -14.3992$, $\bar{\epsilon}_2 = 0.1485$, $\epsilon^* = 0.59396$, and $\rho = 12.1378$. Subsequently, we get that $\epsilon_1 - \epsilon^* = 25.9430 > 0$. Furthermore, equation $\lambda - \epsilon_1 + \epsilon^* e^{\lambda\tau} = 0$ has a unique positive solution $\lambda = 6.9877$. We set $\Psi = 0.49$ and it is easy to get that $\lambda - \rho\Psi = 1.0402 > 0$. Up to now, all conditions in Theorem 3 are verified, which indicates the availability of our AIC. Then, mean square figures of systems (23) and (24) are shown in Figs. 3 and 4. From Fig. 5, we know that drive-response systems (23) and (24) can achieve mean square exponential synchronization.

V. APPLICATION TO COUPLED CHUA'S CIRCUITS

In the past few years, Chua's circuits have been a research focus in many fields. A single coupled Chua's circuit (see [54])

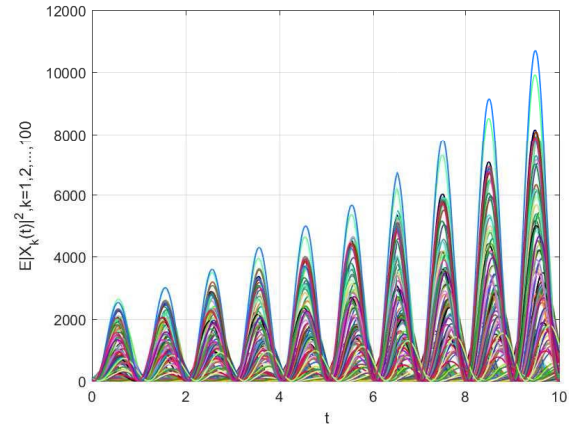


Fig. 3. Mean square figure of drive system (23).

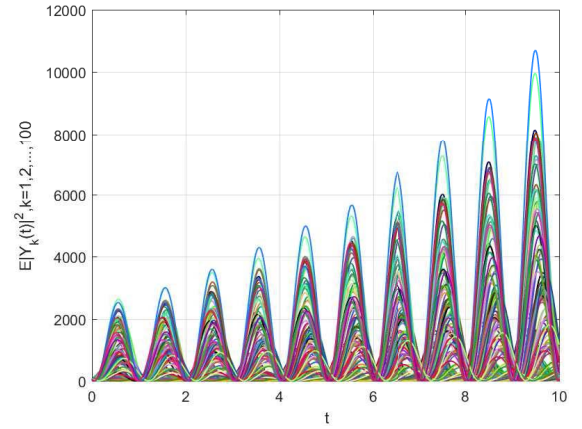


Fig. 4. Mean square figure of response system (24).

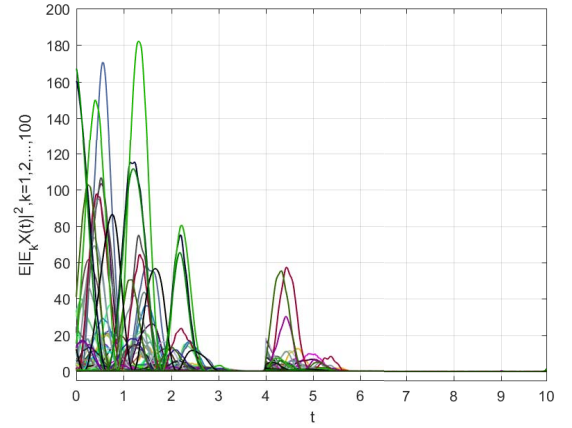


Fig. 5. Mean square figure of error system (26) with control.

is described as follows:

$$\begin{cases} C_1 dv_1(t) = \left[\frac{1}{R}(-v_1(t) + v_2(t)) - \mathcal{D}(v_1(t)) \right] dt \\ C_2 dv_2(t) = \left[\frac{1}{R}(v_1(t) - v_2(t)) + i_3(t) \right] dt \\ L di_3(t) = -(v_2(t) + R_0 i_3(t)) dt \end{cases} \quad (31)$$

where $v_1(t)$ and $v_2(t)$ denote voltages across the capacitors C_1 and C_2 , $i_3(t)$ is the current through the inductances L , R_0 , and R represent the linear resistors, and $\mathcal{D}(v_1(t)) = G_{b1}v_1(t) + 0.5(G_{a1} - G_{b1})(|v_1(t) + 1| - |v_1(t) - 1|)$, in which G_{a1} denotes

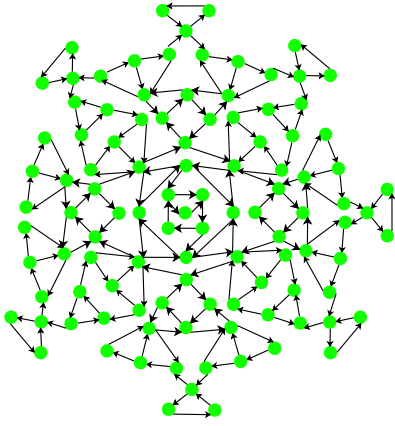


Fig. 6. Coupling structure of 106 coupled Chua's circuits.

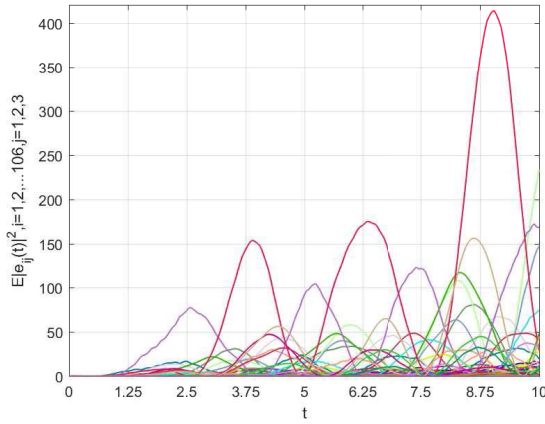


Fig. 7. Mean square figure of error system (35) without control (34).

the slopes of the inner regions while G_{b1} represents the slopes of the outer regions.

Next, we consider N coupled Chua's circuits with time-varying coupling structure and stochastic interference as drive system as follows:

$$\begin{aligned} \begin{bmatrix} \dot{x}_{k1}(t) \\ \dot{x}_{k2}(t) \\ \dot{x}_{k3}(t) \end{bmatrix} &= \begin{bmatrix} -\varsigma & \varsigma & 0 \\ \mu & -\mu & \psi \\ 0 & -v & -z \end{bmatrix} \begin{bmatrix} x_{k1}(t) \\ x_{k2}(t) \\ x_{k3}(t) \end{bmatrix} + \begin{bmatrix} -\omega \mathcal{D}(x_{k1}(t - \tau_1(t))) \\ 0 \\ 0 \end{bmatrix} \\ &+ \sum_{h=1}^N a_{kh}(t) \sin(x_h(t - \tau_3(t)) - x_k(t - \tau_2(t))) \\ &+ rx_k(t - \tau_4(t))\dot{B}(t), \quad k \in \mathcal{N} \end{aligned} \quad (32)$$

where $\varsigma = (1/RC_1)$, $\mu = (1/RC_2)$, $\psi = (1/C_2)$, $v = (1/L)$, $z = (R_0/L)$, and $\omega = (1/C_1)$, $k \in \mathcal{N}$. Besides, $a_{kh}(t)$ denotes the coupling strength, $\tau_2(t)$ represents the self-delay and $\tau_3(t)$ is the neighboring delay, r is a positive constant, and $B(t)$ is a 1-D Brownian motion.

Similarly, the corresponding response system is given by

$$\begin{aligned} \begin{bmatrix} \dot{y}_{k1}(t) \\ \dot{y}_{k2}(t) \\ \dot{y}_{k3}(t) \end{bmatrix} &= \begin{bmatrix} -\varsigma & \varsigma & 0 \\ \mu & -\mu & \psi \\ 0 & -v & -z \end{bmatrix} \begin{bmatrix} y_{k1}(t) \\ y_{k2}(t) \\ y_{k3}(t) \end{bmatrix} + \begin{bmatrix} -\omega \mathcal{D}(y_{k1}(t - \tau_1(t))) \\ 0 \\ 0 \end{bmatrix} \\ &+ \sum_{h=1}^N a_{kh}(t) \sin(y_h(t - \tau_3(t)) - y_k(t - \tau_2(t))) \end{aligned}$$

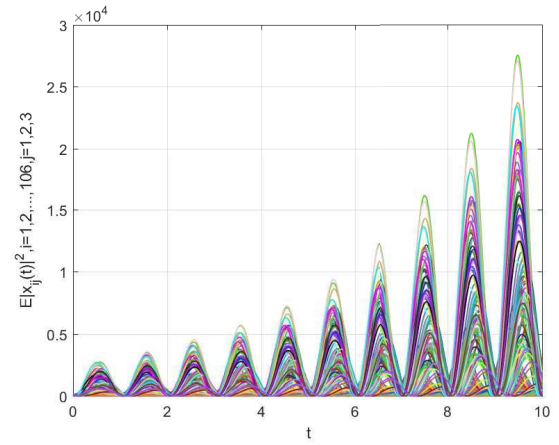


Fig. 8. Mean square figure of drive system (32).

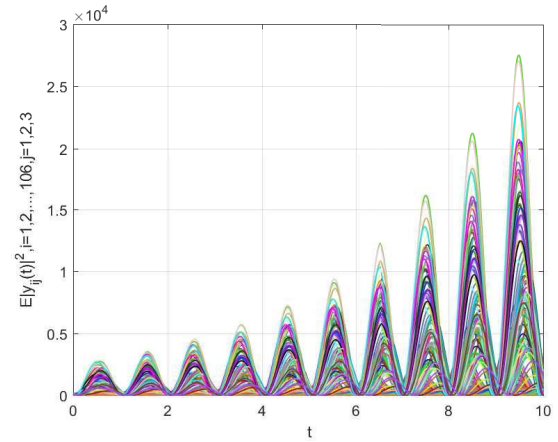


Fig. 9. Mean square figure of response system (33).

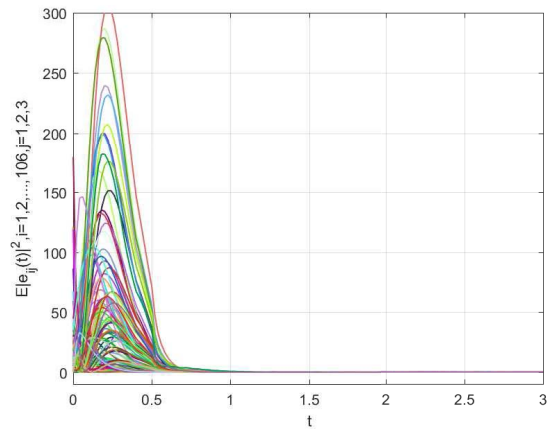


Fig. 10. Mean square figure of error system (35) with control (34).

$$+ U_k(t) + ry_k(t - \tau_4(t))\dot{B}(t), \quad k \in \mathcal{N} \quad (33)$$

where

$$U_k(t) = \Upsilon_k(t)(y_k(t) - x_k(t)) \quad (34)$$

in which

$$\Upsilon_k(t) = \begin{cases} -\Upsilon_k, & t_m \leq t < s_m \\ 0, & s_m \leq t < t_{m+1} \end{cases} \quad m = 0, 1, \dots$$

TABLE I
ELEMENTS OF WEIGHTED MATRIX OF FIG. 1

$a_{1,4}(t)$	$a_{1,5}(t)$	$a_{2,1}(t)$	$a_{3,2}(t)$	$a_{4,3}(t)$	$= 0.8 + e^{-1.7t}$
$a_{5,10}(t)$	$a_{5,12}(t)$	$a_{6,5}(t)$	$a_{6,13}(t)$	$a_{7,6}(t)$	$= 0.8 + e^{-1.4t}$
$a_{7,8}(t)$	$a_{8,14}(t)$	$a_{8,15}(t)$	$a_{9,8}(t)$	$a_{9,16}(t)$	$= 0.9 + e^{-3.2t}$
$a_{10,1}(t)$	$a_{10,9}(t)$	$a_{10,11}(t)$	$a_{11,5}(t)$	$a_{11,18}(t)$	$= 1.0 + e^{-4.0t}$
$a_{11,19}(t)$	$a_{12,6}(t)$	$a_{12,21}(t)$	$a_{12,22}(t)$	$a_{13,7}(t)$	$= 0.1 + e^{-2.5t}$
$a_{13,24}(t)$	$a_{13,25}(t)$	$a_{14,7}(t)$	$a_{14,27}(t)$	$a_{14,28}(t)$	$= 0.1 + e^{-4.3t}$
$a_{15,9}(t)$	$a_{15,30}(t)$	$a_{15,31}(t)$	$a_{16,10}(t)$	$a_{16,33}(t)$	$= 0.1 + e^{-3.7t}$
$a_{16,34}(t)$	$a_{17,11}(t)$	$a_{17,35}(t)$	$a_{18,17}(t)$	$a_{18,19}(t)$	$= 0.2 + e^{-2.1t}$
$a_{19,44}(t)$	$a_{20,12}(t)$	$a_{20,36}(t)$	$a_{21,20}(t)$	$a_{21,22}(t)$	$= 0.15 + e^{-4.6t}$
$a_{22,46}(t)$	$a_{23,13}(t)$	$a_{23,37}(t)$	$a_{24,23}(t)$	$a_{24,25}(t)$	$= 0.7 + e^{-2.9t}$
$a_{25,48}(t)$	$a_{26,14}(t)$	$a_{26,38}(t)$	$a_{27,26}(t)$	$a_{27,28}(t)$	$= 0.9 + e^{-4.3t}$
$a_{28,50}(t)$	$a_{29,15}(t)$	$a_{29,39}(t)$	$a_{30,29}(t)$	$a_{30,31}(t)$	$= 0.5 + e^{-1.8t}$
$a_{31,52}(t)$	$a_{32,16}(t)$	$a_{32,40}(t)$	$a_{33,32}(t)$	$a_{33,34}(t)$	$= 0.6 + e^{-1.1t}$
$a_{34,42}(t)$	$a_{35,34}(t)$	$a_{35,56}(t)$	$a_{36,19}(t)$	$a_{36,59}(t)$	$= 1.0 + e^{-4.4t}$
$a_{37,22}(t)$	$a_{37,62}(t)$	$a_{38,25}(t)$	$a_{38,65}(t)$	$a_{39,28}(t)$	$= 0.5 + e^{-2.9t}$
$a_{39,68}(t)$	$a_{40,31}(t)$	$a_{40,53}(t)$	$a_{41,32}(t)$	$a_{41,55}(t)$	$= 0.2 + e^{-1.8t}$
$a_{42,41}(t)$	$a_{43,17}(t)$	$a_{43,58}(t)$	$a_{44,43}(t)$	$a_{45,20}(t)$	$= 0.2 + e^{-1.2t}$
$a_{45,61}(t)$	$a_{46,45}(t)$	$a_{47,23}(t)$	$a_{47,64}(t)$	$a_{48,47}(t)$	$= 0.6 + e^{-2.0t}$
$a_{49,26}(t)$	$a_{49,67}(t)$	$a_{50,49}(t)$	$a_{51,29}(t)$	$a_{51,70}(t)$	$= 0.4 + e^{-3.7t}$
$a_{52,51}(t)$	$a_{53,54}(t)$	$a_{54,40}(t)$	$a_{54,75}(t)$	$a_{55,75}(t)$	$= 0.1 + e^{-2.1t}$
$a_{55,77}(t)$	$a_{55,78}(t)$	$a_{56,57}(t)$	$a_{57,35}(t)$	$a_{57,80}(t)$	$= 1.0 + e^{-4.4t}$
$a_{58,44}(t)$	$a_{58,82}(t)$	$a_{58,83}(t)$	$a_{59,60}(t)$	$a_{60,36}(t)$	$= 0.9 + e^{-3.5t}$
$a_{60,85}(t)$	$a_{61,46}(t)$	$a_{61,87}(t)$	$a_{61,88}(t)$	$a_{62,63}(t)$	$= 0.7 + e^{-3.7t}$
$a_{63,37}(t)$	$a_{63,90}(t)$	$a_{64,48}(t)$	$a_{64,92}(t)$	$a_{64,93}(t)$	$= 0.3 + e^{-3.4t}$
$a_{65,66}(t)$	$a_{66,38}(t)$	$a_{66,95}(t)$	$a_{67,50}(t)$	$a_{67,97}(t)$	$= 0.4 + e^{-4.7t}$
$a_{67,98}(t)$	$a_{68,69}(t)$	$a_{69,39}(t)$	$a_{69,100}(t)$	$a_{70,52}(t)$	$= 0.6 + e^{-4.6t}$
$a_{70,72}(t)$	$a_{70,73}(t)$	$a_{71,70}(t)$	$a_{71,100}(t)$	$a_{72,71}(t)$	$= 0.7 + e^{-3.1t}$
$a_{73,72}(t)$	$a_{74,53}(t)$	$a_{74,73}(t)$	$a_{75,74}(t)$	$a_{76,55}(t)$	$= 1.0 + e^{-4.0t}$
$a_{76,75}(t)$	$a_{77,76}(t)$	$a_{78,77}(t)$	$a_{79,56}(t)$	$a_{79,78}(t)$	$= 0.2 + e^{-4.1t}$
$a_{80,79}(t)$	$a_{81,58}(t)$	$a_{81,80}(t)$	$a_{82,81}(t)$	$a_{83,82}(t)$	$= 0.8 + e^{-2.4t}$
$a_{84,59}(t)$	$a_{84,83}(t)$	$a_{85,84}(t)$	$a_{86,61}(t)$	$a_{86,85}(t)$	$= 0.1 + e^{-1.4t}$
$a_{87,86}(t)$	$a_{88,87}(t)$	$a_{89,88}(t)$	$a_{90,89}(t)$		$= 0.9 + e^{-2.8t}$
$a_{91,64}(t)$	$a_{91,90}(t)$	$a_{92,91}(t)$	$a_{93,92}(t)$	$a_{94,65}(t)$	$= 0.2 + e^{-1.1t}$
$a_{94,93}(t)$	$a_{95,94}(t)$	$a_{96,67}(t)$	$a_{96,95}(t)$	$a_{97,96}(t)$	$= 0.2 + e^{-4.0t}$
$a_{98,97}(t)$	$a_{99,68}(t)$	$a_{99,98}(t)$	$a_{100,99}(t)$		$= 0.2 + e^{-4.4t}$

TABLE II
ELEMENTS OF WEIGHTED MATRIX OF FIG. 6

$a_{1,4}(t)$	$a_{1,5}(t)$	$a_{2,1}(t)$	$a_{3,2}(t)$	$a_{4,3}(t)$	$= 0.3 + e^{-3.8t}$
$a_{5,10}(t)$	$a_{5,12}(t)$	$a_{6,5}(t)$	$a_{6,13}(t)$	$a_{7,6}(t)$	$= 0.7 + e^{-1.5t}$
$a_{7,8}(t)$	$a_{8,14}(t)$	$a_{8,15}(t)$	$a_{9,8}(t)$	$a_{9,16}(t)$	$= 0.6 + e^{-3.4t}$
$a_{10,1}(t)$	$a_{10,9}(t)$	$a_{10,11}(t)$	$a_{11,5}(t)$	$a_{11,18}(t)$	$= 0.7 + e^{-2.6t}$
$a_{11,19}(t)$	$a_{12,6}(t)$	$a_{12,21}(t)$	$a_{12,22}(t)$	$a_{13,7}(t)$	$= 0.7 + e^{-3.3t}$
$a_{13,24}(t)$	$a_{13,25}(t)$	$a_{14,7}(t)$	$a_{14,27}(t)$	$a_{14,28}(t)$	$= 0.9 + e^{-4.9t}$
$a_{15,9}(t)$	$a_{15,30}(t)$	$a_{15,31}(t)$	$a_{16,10}(t)$	$a_{16,33}(t)$	$= 0.1 + e^{-1.3t}$
$a_{16,34}(t)$	$a_{17,11}(t)$	$a_{17,35}(t)$	$a_{18,17}(t)$	$a_{18,19}(t)$	$= 0.8 + e^{-2.1t}$
$a_{19,44}(t)$	$a_{20,12}(t)$	$a_{20,36}(t)$	$a_{21,20}(t)$	$a_{21,22}(t)$	$= 0.1 + e^{-4.1t}$
$a_{22,46}(t)$	$a_{23,13}(t)$	$a_{23,37}(t)$	$a_{24,23}(t)$	$a_{24,25}(t)$	$= 0.6 + e^{-5.1t}$
$a_{25,48}(t)$	$a_{26,14}(t)$	$a_{26,38}(t)$	$a_{27,26}(t)$	$a_{27,28}(t)$	$= 0.5 + e^{-4.3t}$
$a_{28,50}(t)$	$a_{29,15}(t)$	$a_{29,39}(t)$	$a_{30,29}(t)$	$a_{30,31}(t)$	$= 0.65 + e^{-2.2t}$
$a_{31,52}(t)$	$a_{32,16}(t)$	$a_{32,40}(t)$	$a_{33,32}(t)$	$a_{33,34}(t)$	$= 0.1 + e^{-1.3t}$
$a_{34,42}(t)$	$a_{35,34}(t)$	$a_{35,56}(t)$	$a_{36,19}(t)$	$a_{36,59}(t)$	$= 1.0 + e^{-2.4t}$
$a_{37,22}(t)$	$a_{37,62}(t)$	$a_{38,25}(t)$	$a_{38,65}(t)$	$a_{39,28}(t)$	$= 1.1 + e^{-4.1t}$
$a_{39,68}(t)$	$a_{40,31}(t)$	$a_{40,53}(t)$	$a_{41,32}(t)$	$a_{41,55}(t)$	$= 0.21 + e^{-2.2t}$
$a_{42,41}(t)$	$a_{43,17}(t)$	$a_{43,58}(t)$	$a_{44,43}(t)$	$a_{45,20}(t)$	$= 0.4 + e^{-1.1t}$
$a_{45,61}(t)$	$a_{46,45}(t)$	$a_{47,23}(t)$	$a_{47,64}(t)$	$a_{48,47}(t)$	$= 0.5 + e^{-2.8t}$
$a_{49,26}(t)$	$a_{49,67}(t)$	$a_{50,49}(t)$	$a_{51,29}(t)$	$a_{51,70}(t)$	$= 0.5 + e^{-3.1t}$
$a_{52,51}(t)$	$a_{53,54}(t)$	$a_{54,40}(t)$	$a_{54,75}(t)$	$a_{55,75}(t)$	$= 0.2 + e^{-1.8t}$
$a_{55,77}(t)$	$a_{55,78}(t)$	$a_{56,57}(t)$	$a_{57,35}(t)$	$a_{57,80}(t)$	$= 0.5 + e^{-3.7t}$
$a_{58,44}(t)$	$a_{58,82}(t)$	$a_{58,83}(t)$	$a_{59,60}(t)$	$a_{60,36}(t)$	$= 0.1 + e^{-3.5t}$
$a_{60,85}(t)$	$a_{61,46}(t)$	$a_{61,87}(t)$	$a_{61,88}(t)$	$a_{62,63}(t)$	$= 0.1 + e^{-3.1t}$
$a_{63,37}(t)$	$a_{63,90}(t)$	$a_{64,48}(t)$	$a_{64,92}(t)$	$a_{64,93}(t)$	$= 0.4 + e^{-3.5t}$
$a_{65,66}(t)$	$a_{66,38}(t)$	$a_{66,95}(t)$	$a_{67,50}(t)$	$a_{67,97}(t)$	$= 0.2 + e^{-2.1t}$
$a_{67,98}(t)$	$a_{68,69}(t)$	$a_{69,39}(t)$	$a_{69,100}(t)$	$a_{70,52}(t)$	$= 0.1 + e^{-2.1t}$
$a_{70,72}(t)$	$a_{70,73}(t)$	$a_{71,70}(t)$	$a_{71,100}(t)$	$a_{72,71}(t)$	$= 0.6 + e^{-3.1t}$
$a_{73,72}(t)$	$a_{74,53}(t)$	$a_{74,73}(t)$	$a_{75,74}(t)$	$a_{76,55}(t)$	$= 1.0 + e^{-1.2t}$
$a_{76,75}(t)$	$a_{77,76}(t)$	$a_{78,77}(t)$	$a_{79,56}(t)$	$a_{79,78}(t)$	$= 2.1 + e^{-5.1t}$
$a_{80,79}(t)$	$a_{81,58}(t)$	$a_{81,80}(t)$	$a_{82,81}(t)$	$a_{83,82}(t)$	$= 0.3 + e^{-1.9t}$
$a_{84,59}(t)$	$a_{84,83}(t)$	$a_{85,84}(t)$	$a_{86,61}(t)$	$a_{86,85}(t)$	$= 0.2 + e^{-1.8t}$
$a_{87,86}(t)$	$a_{88,87}(t)$	$a_{89,88}(t)$	$a_{90,89}(t)$		$= 0.5 + e^{-2.2t}$
$a_{91,64}(t)$	$a_{91,90}(t)$	$a_{92,91}(t)$	$a_{93,92}(t)$	$a_{94,65}(t)$	$= 0.8 + e^{-2.1t}$
$a_{94,93}(t)$	$a_{95,94}(t)$	$a_{96,67}(t)$	$a_{96,95}(t)$	$a_{97,96}(t)$	$= 0.8 + e^{-4.5t}$
$a_{94,93}(t)$	$a_{95,94}(t)$	$a_{96,67}(t)$	$a_{96,95}(t)$	$a_{97,96}(t)$	$= 0.8 + e^{-4.5t}$
$a_{98,97}(t)$	$a_{98,105}(t)$	$a_{99,98}(t)$	$a_{100,99}(t)$	$a_{99,68}(t)$	$= 0.2 + e^{-4.4t}$
$a_{101,81}(t)$	$a_{102,86}(t)$	$a_{103,91}(t)$			$= 0.25 + e^{-2.1t}$
$a_{104,96}(t)$	$a_{105,71}(t)$	$a_{106,76}(t)$			$= 0.4 + e^{-3.3t}$

here Υ_k is a positive constant. Let $e_{ki}(t) = y_{ki}(t) - x_{ki}(t)$, $i = 1, 2, 3$. Therefore, we can obtain the following error system:

$$\begin{aligned}
 \begin{bmatrix} de_{k1}(t) \\ de_{k2}(t) \\ de_{k3}(t) \end{bmatrix} &= \begin{bmatrix} -\varsigma & \varsigma & 0 \\ \mu & -\mu & \psi \\ 0 & -\nu & -z \end{bmatrix} \begin{bmatrix} e_{k1}(t) \\ e_{k2}(t) \\ e_{k3}(t) \end{bmatrix} \\
 &+ \begin{bmatrix} -\omega \tilde{\mathcal{D}}(e_{k1}(t - \tau_1(t))) \\ 0 \\ 0 \end{bmatrix} + \sum_{h=1}^N a_{kh}(t) \tilde{\sin} \\
 &\times (e_h(t - \tau_3(t)) - e_k(t - \tau_2(t))) + U_k(t) \Big] dt \\
 &+ re_k(t - \tau_4(t)) dB(t), \quad k \in \mathcal{N} \quad (35)
 \end{aligned}$$

where $\tilde{\mathcal{D}}(e_{k1}(t - \tau_1(t))) = \mathcal{D}(y_{k1}(t - \tau_1(t))) - \mathcal{D}(x_{k1}(t - \tau_1(t)))$, $\tilde{\sin}(e_h(t - \tau_3(t)) - e_k(t - \tau_2(t))) = \sin(y_h(t - \tau_3(t)) - y_k(t - \tau_2(t))) - \sin(x_h(t - \tau_3(t)) - x_k(t - \tau_2(t)))$.

Our main purpose is to utilize AIC to achieve mean square exponential synchronization of drive-response systems (32) and (33). Let $N = 106$. The coupling structure of 106 coupled Chua's circuits is shown in Fig. 6. Moreover, let $\varsigma = 9.5$, $r = 0.2$, $\mu = 0.8$, $\psi = 1.07$, $\omega = 9.3$, $G_{b1} = -0.7349$, $G_{a1} = -1.3955$, $\nu = 16.4834$, and $z = 0.1377$. Meanwhile, the elements of weighted matrix $A(t) = (a_{kh}(t))_{106 \times 106}$ are shown in Appendix B.

The mean square figure of error system (35) without AIC (34) is shown in Fig. 7. We can easily obtain that systems (32) and (33) cannot achieve synchronization. Hence, AIC is necessary to be introduced. For simplicity, let $\Upsilon_1 = \Upsilon_2 = \dots = \Upsilon_{106} = 20.6$. The figures of drive-response systems (32) and (33) are presented in Figs. 8 and 9. Then, the mean square figure of error system (35) with control is shown in Fig. 10. Based on Fig. 10, we can easily derive that systems (32) and (33) can achieve mean square exponential synchronized under AIC (34).

VI. CONCLUSION

In this paper, an aperiodically intermittent controller was used to study the PMES of TCDNMTDs. Time delays, including internal time-varying delays and coupling time-varying delays were taken into consideration and they were different from each other. Meanwhile, the synchronization problems of SKOs and coupled Chua's circuits were also considered. In our future work, we will consider two topics: 1) we will use pinning aperiodically intermittent controller to study the synchronization of stochastic complex networks, which is a challenging and interesting problem and 2) the restrictive

condition on the time-varying coupling weights of complex networks will be further relaxed and weakened.

APPENDIX A

See Table I.

APPENDIX B

See Table II.

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